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PowerExchange for DB2 for i5/OS CDC: Using the Delete Journal Receiver Exit Program

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Abstract

This article descibes key considerations for PowerExchange use of the Delete Journal Receiver exit program for DB2 for i5/OS change data capture (CDC). PowerExchanges uses this exit program to prevent the deletion of journal receivers while CDC sessions read change data from them.

Supported Versions

- PowerExchange 10.0 or later

Table of Contents

Overview. 2
Usage Considerations. 2

Overview

When you install PowerExchange for DB2 for i5/OS CDC, PowerExchange registers the Delete Journal Receiver exit program at the QIBM_QJO_DLT_JRNRCV exit point. PowerExchange uses this exit program to prevent the deletion of journal receivers while CDC sessions read them for change capture. The exit program prevents deletion by locking the journal receivers.

For PowerExchange to use the Delete Journal Receiver exit program, you must specify AS4JRNEXIT=Y in the AS4J CAPI_CONNECTION statement in the DBMOVER configuration member of the *datalib* /CFG file. With this setting, the exit program locks the journal receiver that a CDC session is reading by creating a lock record in the PWXJRNLCKP lock file in the CONDLIB library. If you issue a DLTJRNRCV command against the locked journal receiver, the exit program checks the lock file for records that reference the journal receiver. If the exit program finds a lock record for the journal receiver, the exit program returns a response that indicates not to delete the journal receiver. When PowerExchange switches to the next journal receiver on the chain, PowerExchange removes the lock record for the journal receiver that was last read and adds a lock record for the journal receiver to be read next.

Usage Considerations

If you plan to use the Delete Journal Receiver exit program for DB2 for i5/OS CDC, read the following usage considerations:

- The Delete Journal Receiver exit program must be able to access the PWXJRNLCKP lock file to read the lock records. Otherwise, the exit program cannot prevent the deletion of journal receivers.
- The Delete Journal Receiver exit program runs under the QUSER user profile by default. During installation, PowerExchange sets *USE authority to PUBLIC on PowerExchange objects to avoid access issues that are related to exit program use of the QUSER profile. If the security requirements at your site do not allow use of PUBLIC authority on PowerExchange objects, the exit program might not be able to access the PWXJRNLCKP file under the QUSER profile. In this case, the exit program might delete journal receivers that are in use by CDC sessions. To prevent this problem, use one of the following strategies:
  - If the owner of the PowerExchange objects has access to the PWXJRNLCKP lock file, issue the following command to change the attributes of the DTLRCVRX exit program from *USER to *OWNER:

    ```
    CHGPGM PGM(CONDLIB/DTLRCVRX)  USRPRF(*OWNER)
    ```
  - When you register the PowerExchange Delete Journal Receiver exit program at the QIBM_Q10_DLT_JRNRCV exit point, specify the user profile under which the exit program runs as program data. Usually, you register the exit program during installation by running the PowerExchange CRTPWXENV command with the

JRNEXTSEQ parameter. To register the exit program or specify a user profile in the program data after installation, you can use the DB2 for i5/OS ADDEXITPGM command. When you specify the command, include the length of data and program data in the PGMDTA keyword. For example, the following ADDEXITPGM command registers the exit program as program number 3131 with a data length of 10 and a profile name of NEWUSER:

```
ADDEXITPGM EXITPNT(QIBM_QJO_DLT_JRNRCV)
           FORMAT(DRCV0100) PGMNBR(3131)
           PGM(JBBV861CL/DTLRCVRX) REPLACE(*YES)
           PGMDTA(*JOB 10 NEWUSER
```

Note: When you include REPLACE(*YES) in this command, the command adds the exit program if it was not added at installation, or the command updates the program data if the exit program is already registered at the exit point.

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